Practice 1: Installation of R Program and Import Packages

1. **Aim**: To install R and import essential packages in R.
2. **Procedure**:
   - Install R from the official website: https://cran.r-project.org/
   - Install the required package(s) using `install.packages()` in R.
   - Use `library()` to load the package.
3. **Simple Program**:

```
r
Copy
# Install a package (if not installed)
install.packages("ggplot2")

# Import the package
library(ggplot2)
```

4. **Input**: None
5. **Output**: No output, but you can check if the library is loaded by using a function like `ggplot2::ggplot()`.

Practice 2: Implementation of R Program – Basic

1. **Aim**: To understand basic R syntax and printing output.
2. **Procedure**:
   - Create simple variables.
   - Print their values using `print()` or `cat()`.
3. **Simple Program**:

```
r
Copy
# Basic program to print Hello World
message <- "Hello, World!"
print(message)
```

4. **Input**: None
5. **Output**:

```
csharp
Copy
[1] "Hello, World!"
```

Practice 3: Implementation of R Program – Basic (Addition)

1. **Aim**: To perform basic arithmetic operations.
2. **Procedure**:
   - Define two variables.
   - Add them using the `+` operator.
   - Print the result.

3. **Simple Program**:

```
r
Copy
# Adding two numbers
num1 <- 5
num2 <- 3
sum <- num1 + num2
print(sum)
```

4. **Input**:
   - `num1 = 5`
   - `num2 = 3`
5. **Output**:

```
csharp
Copy
[1] 8
```

Practice 4: Implementation of Data Types in R

1. **Aim**: To understand and implement various data types in R.
2. **Procedure**:
   - Create variables of different types: numeric, character, logical, and complex.
   - Print the type using `typeof()`.
3. **Simple Program**:

```
r
Copy
# Creating variables of different types
num <- 10
text <- "Hello"
is_true <- TRUE
complex_num <- 1 + 2i

# Print types
print(typeof(num))         # Numeric
print(typeof(text))        # Character
print(typeof(is_true))     # Logical
print(typeof(complex_num)) # Complex
```

4. **Input**: None
5. **Output**:

```
csharp
Copy
[1] "double"
[1] "character"
[1] "logical"
[1] "complex"
```

Practice 5: Implementation of Control Statements in R

1. **Aim**: To implement basic control structures (if-else).
2. **Procedure**:
   - Use `if`, `else`, and `else if` for decision-making.
3. **Simple Program**:

```
r
Copy
# Check if the number is positive or negative
num <- -5

if (num > 0) {
  print("Positive")
} else if (num < 0) {
  print("Negative")
} else {
  print("Zero")
}
```

4. **Input**: `num = -5`
5. **Output**:

```
csharp
Copy
[1] "Negative"
```

Practice 6: Implementation of Looping Statements

1. **Aim**: To implement basic looping structures like `for` and `while` loops.
2. **Procedure**:
   - Use a `for` loop to print numbers from 1 to 5.
   - Use a `while` loop to print numbers from 6 to 10.
3. **Simple Program**:

```
r
Copy
# For loop example
for (i in 1:5) {
  print(i)
}

# While loop example
i <- 6
while (i <= 10) {
  print(i)
  i <- i + 1
}
```

4. **Input**: None
5. **Output**:

```
csharp
Copy
[1] 1
[1] 2
[1] 3
```

```
[1] 4
[1] 5
[1] 6
[1] 7
[1] 8
[1] 9
[1] 10
```

Practice 7: Implementation of Decision Tree and KNN in R

1. **Aim**: To implement decision trees and KNN algorithms for classification tasks.
2. **Procedure**:
   - Install and load the necessary package `rpart` for decision tree and `class` for KNN.
   - Use sample data for training.
3. **Simple Program for Decision Tree**:

   ```
   r
   Copy
   # Load necessary libraries
   install.packages("rpart")
   library(rpart)

   # Sample data
   data(iris)

   # Create a decision tree
   model <- rpart(Species ~ Sepal.Length + Sepal.Width, data = iris)

   # Print the model
   print(model)
   ```

4. **Input**: `iris` dataset (comes preloaded in R)
5. **Output**: Decision tree model summary.

Practice 8: Implementation of Naïve Bayes

1. **Aim**: To implement the Naïve Bayes classification algorithm.
2. **Procedure**:
   - Install and load the `e1071` package.
   - Use `naiveBayes()` to create a model.
3. **Simple Program**:

   ```
   r
   Copy
   # Install and load the e1071 package
   install.packages("e1071")
   library(e1071)

   # Sample data
   data(iris)
   ```

```
# Apply Naive Bayes
model <- naiveBayes(Species ~ Sepal.Length + Sepal.Width, data =
iris)

# Print model summary
print(model)
```

4. **Input**: `iris` dataset
5. **Output**: Naïve Bayes model summary.

Practice 9: Implementation of Random Forest in R

1. **Aim**: To implement a random forest model for classification.
2. **Procedure**:
   - Install and load the `randomForest` package.
   - Create and train a random forest model.
3. **Simple Program**:

```
r
Copy
# Install and load the randomForest package
install.packages("randomForest")
library(randomForest)

# Sample data
data(iris)

# Create a Random Forest model
model <- randomForest(Species ~ Sepal.Length + Sepal.Width, data =
iris)

# Print the model
print(model)
```

4. **Input**: `iris` dataset
5. **Output**: Random Forest model summary.

Practice 10: Implementation of K-means in R

1. **Aim**: To implement the K-means clustering algorithm.
2. **Procedure**:
   - Use `kmeans()` to perform clustering on the data.
   - Plot the clusters.
3. **Simple Program**:

```
r
Copy
# Sample data
data(iris)
```

```
# K-means clustering
set.seed(10)
clusters <- kmeans(iris[, 1:4], centers = 3)

# Print clustering results
print(clusters)
```

4. **Input**: `iris` dataset
5. **Output**: K-means clustering results, including cluster centers.